# Table-top lab · guide

How Animals Get Ready for Winter

Wood frogs: ice forms outside cells while liver glucose protects them; on thawing the heart returns first. Chickadees cache food and use the hippocampus to find it; one study found more new neurons there in fall, but later studies disagree and the cells seem to replace old ones, so we never say they grow bigger brains. In most small mammal hibernators, hibernation is long torpor bouts with brief rewarming, often fuelled by brown fat; a bear cools only a few degrees. With no brood, honeybee clusters keep the core above about 64 F. Day length is thought to trigger the hare molt (about 40 days). About 6 in of snow keeps the subnivean zone near 32 F. Woolly bears make glycerol; we skip the stripe folklore. Watch wildlife from a distance; use warm, not hot, water.

### Try 1 · How can a wood frog freeze and live?

Draw a big circle for one frog cell. Draw ice crystals only outside the circle. Draw sugar dots near the cell. Tell how the sugar helps keep the cell safe while ice forms around it.

**Look for:** When a wood frog starts to freeze, its liver pours out a sugar called glucose. Ice forms in its body, but outside its cells, and the glucose helps protect the cells. When the frog thaws, its heart is the first part to start again.

### Try 2 · How does a chipmunk save energy all winter?

Sit very still and count your breaths for 30 seconds. Then march in place for 30 seconds and count again. Tell which time used more energy. In torpor, an animal slows its body way down to save energy.

**Look for:** A chipmunk is a small hibernator. Its body gets much colder on purpose. This is called torpor, and it saves energy. For most small hibernators like this one, hibernation is many long torpor stretches in a row, with short warm-ups between them. To help warm back up, many mammal hibernators burn a special fat called brown fat.

### Try 3 · How do honeybees stay warm without hibernating?

Fill two jars with warm tap water and close the lids. Wrap one jar in a thick towel or some socks, like the outside bees. Leave the other jar bare. Wait 20 minutes, then touch both jars and tell which one stayed warmer.

**Look for:** Honeybees do not hibernate. They pack into a ball called a winter cluster. The outside bees make a blanket. The inside bees shiver their flight muscles without moving their wings, which makes heat. When there are no baby bees in the hive, the middle of the cluster is kept above about 64 degrees Fahrenheit.

### Try 4 · Why can a mouse stay warmer under the snow?

Put two kitchen thermometers in the freezer. Bury one in a thick pile of cotton balls, like a snow blanket. Leave the other one bare. After 20 minutes, read both and tell which one is warmer.

**Look for:** The space between the ground and the snow is called the subnivean zone. About six inches of snow works like a blanket, so this space stays near 32 degrees Fahrenheit even when the air above is much colder. Voles and mice tunnel and eat there, and small air holes let carbon dioxide out.